2024-04-09: Year Two Meeting Minutes

Presents

Room 3000: Guilhem Barruol, Wayne Crawford, Richard Dreo, Veronique Farra, Frederic Guattari, Jean-Paul Montagner, Eleonore Stutzmann, Zongbo Xu

On line: Mohammad-Amin Aminian, Martin Schimmel, Fabrice Ardhuin, Spahr Webb

Program

Work Package Updates

WP1: Administration

- The website has been updated for the Budget page, new available data, and a page (unfinished) on the Noise Reduction Challenge
- Budget
  - We are halfway through the projet, and all temporary positions have been filled
  - 1 internship (Wassim Seksaf, Mediterranean IG waves) was paid for
  - Travel has been spent or is set aside for:
    - WP1 meetings (~6k€ including this meeting)
    - 1 EGU 2024 (~2k Crawford)
    - 2 AGU 2024 (~5,5k€ Crawford and Xu)
  - Remaining:
    - 29k€ (80%) travel
    - 5k€ (70%) internships*
    - 12k€ (100%) publication charges
    - ~28k€ from IPGP rotational OBS budget, which will be divided as:
      - 5k new rotational work (complementing 33k€ from eXail)

        - 15k for Noise Removal Challenge
        - 12k for internships
- Noise Reduction Challenge page
    - Includes ARC-EN-SUB data and processing
        - Has a `tiskitpy` example
        - **Needs a Bruit-FM toolbox example**
        - **Need to specify submission formats and deadlines**
        - **Can we create/add synthetic data?**

> **Eleonore:**
>   Can you modify noise reduction challenge figures to show what has been thrown away?
>
> **Richard**
>   Need to update the "Recommended BRUIT-FM datasets" table so that it reflects what Richard has downloaded
>
>   Japanese deployments aren't shown on the map
>
>   Most networks downloaded by Richard are > 3 months
>
> **Wayne**
>   Maybe should update "Recommended" table to say which are downloaded and make a separate table of what is downloaded but not on recommended list
>
>   **"Depth" on Richard's Network Stations tables should be written as "Elevation"**
>
> **Zongbo**
>   "NM" network looks like it is actually on land (New Madrid)
>
> **Wayne**
>   If you look on FDSN site, NM network is clearly on land and with a station distribution unlike that seen on the BRUIT-FM map, plus station names do not start with "NM"
>
>   **Is this a Japanese deployment** (it is near to Japan)?

WP2: Full Seafloor Spectrum

The inventory maps have been updated.

If you replace `inventory` with `downloaded` in the URL, you will see all of the datasets that Richard Dreo has downloaded, from which he can send you datasets that interest you

Richard supplied data to Zongbo. Zongbo has not verified

> **Zongbo**
> : Would be useful to have 1 Hz data [and other unheard...]
>
> **Wayne**
> : Most OBSs don't offer 1 Hz ouput. Only Woods Hole puts it out, because their data logger (Q330S) records it naturally.
>
> **Richard**
> : I do downsample using obspy
>
> **Wayne**
> : Better to use the tiskitpy Decimator module, which uses known decimation filters and puts their information into a new metadata file.
> : Uses SAC filters (~8-bit BB rejection), might want to use a more modern decimation filter in the future
>
> **Eleonore**
> : Be very careful of obspy downsampling
>
> **Richard**
> : I haven't seen any change in spectra after obspy decimation.
>
> **Wayne**
> : **We should talk about decimation together**
> : And maybe we should make some scripts/codes to simplify copying selected data from a big SDS database to another disk

WP3: The Generation of Global Seismic Noise

webpage

Very happy to have hired Zongbo in October of last year. He will present his work, which I think is going well.

Wayne
: Seksaf presentation is relevant to WP3 (more than to WP4)

WP4: Signal Separation and Noise Removal

webpage

Task 4.1: Reducing noise using a rotational seismometer

- Exail (ex-iXblue) withdrew from the consortium
  - No funds spent
  - We have requested ANR to redistribute the funds to IPGP, approved in principal, but waiting on paperwork
  - Big problem for Ifremer, because they haven't been paid for over a year (no consortium agreement)

Task 4.2: Signal processing techniques for signal separation and noise removal

- Simon Rebeyrol (Ifremer postdoc) developed the BRUIT-FM toolbox, article submitted and in second round of reviews. Simon left Ifremer in August 2023 but is finishing up the modification in his spare time.
- The tiskitpy module is validated.
  - Code available on github and via pip
  - Documentation available on readthedocs
  - There are classes for `SpectralDensity`, `Decimator`, `DataCleaner`, `CleanRotate`, `ResponseFunctions` and `TimeSpan`

- Noise Removal Challenge concept developed
- **To Do:**
  - **Improve the conventional transfer function method**
  - **Signal separation based on adaptive template separation**
  - **Broad source separation methods relying on very-limited modelling assumptions**

Task 4.3: Separating seismological and biological signals

- "High risk, high reward"
- Was supposed to be Simon Rebeyrol's 2nd project but he left before he could start it and Ifremer is reluctant to hire a 2nd postdoc.

WP5: The Seafloor Soundscape

webpage

Two main projects this year: whale songs (Richard) and cryoseismology (Guilhem). Both will be presented today.

Challenges and Opportunities

ixBlue/eXail's departure

- No change to IFREMER status
- Much delay in progress (and Ifremer payments)
- Available funds (66,6k€):
  - 28k€ IPGP
  - 35,2k€ from eXail (39,5 total, of which 4,3 go to IPGP overhead)
  - 3,4k€ Ifremer
- Proposed to ANR to use unused eXail + Ifremer funds to
  - Design (but not build) a better BBOBS rotational seismometer (39,7k€)
    - Source for a future dedicated ANR proposal
  - Fund a “noise removal challenge” (14,9k€)
  - Increase the number of internships (12k€)

Departure of Grenoble GIPSA-lab colleagues

On 5 June 2023 (the eve of the first-year meeting), Jerome Mars and Michel Olivier informed Wayne of their intention to leave the project. This is a major loss. We understand the difficulty of participating in too many projects without significant funding, but Jerome and Michel's signal processing techniques should be an important part of the development of new noise separation techniques.

Wayne proposes to offer an internship, in collaboration with Jerome and Michel, to apply their techniques to seafloor data, perhaps the Noise Challenge data. Moreover, we propose to increase the internship budget so that any BRUIT-FM collaborator can easily direct an intership project related to BRUIT-FM.

> **Jean-Paul**
>
> What happened to rotational seismometers developed by iXblue?
>
> **Frederic**
>
> Not available from iXblue, not allowed to make outside

Science Talks

Cryoseismicity of Astrolabe glacier

G Barruol

- Slideshow (in French)
- Recording

Seafloor Rotational Seismology

F Bernauer, W Crawford and F Guattari

- Slideshow 1 (Felix+Wayne)
- Recording 1
- Slideshow 2 (Frederic) (minus 4 redacted slides)
- No Recording of the 2nd slideshow

MAAGM will propose a pasive ring-laser gyro, with improved optical readout displacement technologies allowing 10^{-13} m/VHz in a transportable system. Goal is to obtain 10^{-9} rads^{-1}Hz$^{-1/2}$ with acceptable dimensions and power consumption for seafloor deployments.

> **Jean-Paul**
> The EMSO ERIC test site is very shallow, isn't it?
>
> **Felix**
> Yes, 20m
>
> **Eleonore**
> Have you thought of the ANTARES (EMSO-Ligure) site? It is deeper and in a more seismically active region

How to build/finance a watertight container?

> **Frederique**
> We would be ready a finance this part ourselves. We have a contact that can make a shallow-water pressure case for < 10k€.
>
> **Wayne**
> I think this is the best option for the EMSO ERIC test site but a deeper site, such as EMSO-Ligure, would require a much stronger pressure case. We have bigger pressure cases that go to 6000 m (BBOBS pressure spheres), but they are provided in priority to users of the SMM instrument park (we have no spares, only the ones on our BBOBSs)
>
> **Felix**
> We need no new technology for first EMSO deploy: all the instrumentation is existing and tested
>
> **Jean-Paul**
> Could be a good way to use BRUIT-FM money: to support this experiment

Side question about the originally-planned BBOBS Blueseis-1C integratino

> **Jean-Paul**
> Why didn't the original rotational OBS have 3 rotational components
>
> **Wayne**
> It would require a new, bigger sphere, which was outside of our budget
>
> **Eleonore**
> Why weren't your modeled rotational seismometers circular?
>
> **Wayne**
> Couldn't, given the physical constraints of the existing BBOBS, its gimbals and the sphere center ring
>
> Felix, why are the noise curves for all of the rotational seismometers increasing from a straight line at low frequencies?
>
> **Felix**
> Has to due with tuning?

MAAGM presentation

F Guatari

- Spent 6 months preparing the company
- Aiming for 10^{-9} rad/s/Hz sensitivity and portability
  - 10^{-10} would be possible for a non-portable laboratory sensor
- BlueSeis 3A was 2x10^{-8}
- Aim for 5W for 3 axes

> **Guilhem**
> We want to develop a cheap North finder (1°, <10k€) for land station orientation. These new developments should allow this.
>
> **Eleonore**
> Would they be adaptable to boreholes?
>
> **Frederique**
> 30 cm canister size

Ocean bottom microseism observation and modelling

Zongbo Xu

- Slideshow
- Recording

> **Wayne**
> How do motion changes with depth relate to Rayleigh versus Stonely ver Scholté waves?
>
> How do OBS microseism amplitues compare to standard "admittance" (is the motion suprising? Or is the pressure?
>
> How much is non-propagating (local)?
>
> What are your receiving coefficient maps based on?

Note that Fabrice Ardhuin claims to have advanced on reflection coeffs

Indian Ocean seafloor shear velocity imaging using compliance

Mohammad-Amin Aminian

- Slideshow
- Recording

Trajectory estimate of baleen whales using a single OBS

Richard Dreo

- Slideshow
- Recording
- Give him LSVQ *data?*

> **Wayne**
> Does using a more physical ray-tracing model improve the result?
>
> **Richard**
> No significant difference for travel-times
>
> **Wayne**
> Could make a difference for identifing which arrival is possible.
>
> **Wayne**
> In Diapo 4, Is misalignment betwen arrival pairs due to non-constant velocity?

IG wave climate on the Mediterranean Sea

Wassim Seksaf, presented by W Crawford

M2 internship with W Crawford and Xavier Bertin

- Presentation
- No Recording (Zoom error)

> **Wayne**
> What can we do now to make this publishable?
>
> **Eleonore**
> Measure travel times and beamform
>
> Focus on storm, track it and calculate
>
> Propagate as surface wave and gets to coast
>
> **Jean-Paul**
> Microseisms: Atlantic vs Mediterrannean, compare with Gualtari paper conclusions.

2024 projects

Rotational OBS

> **Frederic**
> : Goal is how to create the best seismometer, need information about design and performance and dimensions. ?how to install into OBS (with or w/o cable, etc)
> : End of 2026, also commitments w/Felix for prototpye in that time scale. Limit risk on manufacturing where, if delayed, we still get report.
>
> **Felix**
> : EMSO-ERIC call 2024-26, including demonstration experiment
> : Need casing for 20m depth
>
> **Frederic**
> : If we are successful to have a sensor for 2026, we will invest in the pressure case.
>
> **Group**
> : High risk, high reward project. Also cements IPGP collaboration.
> : Helps us by keeping close, helps him through OBS expertise.

Decision:
- **Go forward with funding MAAGM to collaborate on design for new 6C BBOBS**
- Look into how to fund (tender? Jeune Entreprise Innovante?)
- Contours of collaboration
- Responsabilties of each
- Deliverables
- Timelines
- Participation in expts w/Felix?

Noise Reduction Challenge

Idea is to show how we reduce noise and calculate compliance on 1/2 datasets, ask for community to do same. We will compare and make a community paper.

- Datasets
  - Currently one site from ARC-EN-SUB experiment
  - Add a synthetic dataset? (would allow us to better evaluate compliance results)
- Define results submission format, prepare example
  - Cleaned data
  - Estimated compliance
-

> **Eleonore**
> : Base on InSight challenge?
> : Contributors must provide codes (if want to be on paper)
> : Provided ocean bottom spectrogram location (2 days). Was too short
> : Should show selected data in Noise Challenge waveform figures for

Synthetic Data

> **Wayne**
>
> should include noise on pressure data, which invalidates simple multiplication by coherence
>
> **Spahr**
>
> Edge of compliance band could also have spectral leakage problem
>
> Put the seismometer on a block of foam and see how well the tilt removal works.

Will make presentations at 2024 AGU and 2025 EGU

Currently 15k€ budget (assuming we recover eXail money)

Seafloor spectral noise levesl

- Like Peterson et al for seismology
- Based on Brown et al (CTBT) methodology?
- Use all recovered data
- Need to establish a protocol to validate instrument response
  - Can use Deng et al. 2022 pressure-acceleration relation at frequencies below 1500/4*H*
  - Use p/u = rho*c for higher frequencies
  - Compare noise curves for sites near to each other
  - Divide results into four boxes:
    - Confirmed
    - Corrected (empty at this stage)
    - Incorrect
    - Bad: for spectra that have no resemblance to the expected form
  - If the number/coverage of confirmed is insufficient, correct as many invalid as possible, put them into "Corrected" box

Verification of signals

- Comparison with nearby stations
- Comparison with predicted p/a (Rayleigh waves, P wave arrivals...)

Correction of signals

- If not enough verified in first round
- Uncorrectable go into 3rd box: junk

Evaluation of results

- Overall spectral bounds
- Bounds based on site parameters
  - seafloor depth, type
  - proximity to marine traffic highways
  - proximity to windmills?
  - Latitude?
  - ...

To Do

(Summary, copied from previous sections)

Short-term

Wayne

- [ ] Modify Noise reduction challenge figures to show what data was thrown away
- [ ] Establish method for funding MAAGM for collaborative study of 6C BBOBS design
    - [ ] Define contours of collaboration

Richard

- [ ] Update website's "Recommended BRUIT-FM datasets" table so that it reflects what Richard has downloaded
- [ ] Change "Depth" to "Elevation" on Network Stations tables
- [ ] Add Japanese deployments to the map
- [ ] Figure out what is wrong with the "NM" network (FDSN indicates it's an on-land "New Madrid" (USA) network, but there are stations in the ocean: do they belong to another network?
- [ ] Discuss decimation method with Wayne

2024-5

Noise reduction Challenge

- [ ] Add Bruit-FM toolbox example
- [ ] Add specification of submission formats and deadlines
- [ ] Create/add synthetic data? (who?)